installed on 1,140 km of routes in the Secunderabad-based South Central Railway and the plan was to roll it out on 35,000-plus km of the network across India. "The way we took up electrification of the rail network, in mission mode, we will do that with the rollout of Kavach as well," he said. However, as of July, Kavach stands implemented only on 1,465 km of the total route length of 68,000 km—a progress of just over 300 km in two years and still limited to the South Central Railway. Similarly, from 90 engines, Kavach is present in 144 engines now—less than 1 per cent of the total 15,200 engines.

Two of the accidents that have taken place in the past two years have been collisions, killing over 20 people. Ironically enough, one of them was in Vizianagaram in Andhra Pradesh, where Kavach is being tested. The other was in West Bengal, the latest one. However, Indian Railways has maintained that the tragic triple train crash near Balasore in Odisha last year, which claimed around 300 lives, was caused by wilful tampering of assets and sabotage, factors that Kavach would not have prevented.

Documents available with INDIA TODAY show that the Commissioner of Railway Safety (CRS), the statutory body under the civil aviation ministry that probes train accidents, had recommended—in no uncertain terms—Kavach, in some shape or form, to save lives, even in stretches not yet included in its rollout plan. The CRS investigation report on the October 2023 Vizianagaram collision recommended that the anti-collision features of Kavach be fitted in all locomotives, even in non-Kavach areas. "To use non-signalling-based anti-collision features of Kavach, it must invariably be fitted on each and every locomotive/ cab across Indian Railways even in non-Kavach territory," the report said.

This is not a difficult job. The Kavach system has two parts. One is the telecom-based signal system that continuously keeps a tab on the movement of Kavach-enabled trains. This requires installing equipment along tracks, telecom towers near stations etc. The other is the hardware in the engine that can show the signals inside the driver's cab and even automatically apply brakes in case it jumps the signal. While the first is a time-consuming job, the installation of the second is relatively easier. Even so, the process is taking an inordinate amount of time.

TWO YEARS AGO, KAVACH WAS PRESENT IN 90 ENGINES. NOW, IT IS IN 144 ENGINES. THIS INDICATES ISSUES BEYOND MERE PROJECT COMPLEXITIES, ARGUE EXPERTS

This slow progress points to issues beyond mere project complexities, argue those with decades of experience in the railway system. Following the CRS recommendation, sources say, there is agreement within the establishment about fitting all new locomotives with Kavach, even for non-Kavach stretches. "In case the ministry feels that the system has not matured enough to be foolproof yet," says Sudhanshu Mani, former general manager of the Integral Coach Factory in Chennai, "why not simultaneously install the European Train Control System (ETCS) Level 2 on some important stretches until we have full confidence in Kavach and it fully matures?"

Kavach is not just any technology; it represents a significant leap for Indian Railways, marking a move towards sophisticated safety measures that have been standard in global rail systems since the 1980s. The journey towards Kavach began in earnest with the approval of the Train Collision Avoidance



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